

Q. No.	Questions
UNIT 1	SECTION 1
1	Two cards are drawn one after another without replacement from a pack of 52 cards. Find the probability that (a) Both are Kings. (b) One is a King and the other is a Queen. (c) Neither card is a face card. (a) Both Kings $P = \frac{4}{52} \times \frac{3}{51} = \frac{1}{221}$ (5 Marks)
2	Three archers P, Q, and R independently shoot at a target. The probability of hitting the target is $P=3/5$, $Q=2/5$ and $R=4/5$. Find the probability that a). None of them hits the target. b). At least one of them hits the target. c). Exactly one of them hits the target. Let $P(P) = \frac{3}{5}, \quad P(Q) = \frac{2}{5}, \quad P(R) = \frac{4}{5}$ Miss probabilities $P(P') = \frac{2}{5}, \quad P(Q') = \frac{3}{5}, \quad P(R') = \frac{1}{5}$ (i) None hits Since the events are independent, $P(\text{None}) = \frac{2}{5} \times \frac{3}{5} \times \frac{1}{5} = \frac{6}{125}$ (5 Marks)
3	In a class of 50 students, 30 study Mathematics, 25 study Physics and 10 study both Mathematics and Physics. If one student is selected randomly, find the probability that the student (i) studies Mathematics (ii) studies Physics (iii) studies either Mathematics or Physics (iv) studies neither Mathematics nor Physics.

	Total students = 50 (i) $P(M) = \frac{30}{50} = \frac{3}{5}$ (ii) $P(P) = \frac{25}{50} = \frac{1}{2}$ (5 Marks)												
4	(i) A university has 60% undergraduate and 40% postgraduate students. Scholarship rates are 20% and 35% respectively. A scholarship holder is selected. Calculate the probability that the student is postgraduate. Hint: Bayes' theorem: $P(PG S) = \frac{P(S PG)P(PG)}{P(S)}$ (3 Marks) (ii) In a group of equal number of men and women, 10% men and 45% women are unemployed. Compute the probability that a person selected at a random is employed Hint: $P(A) = P(B) \cdot (A/B) + P(C) \cdot P(A/C)$ (2 Marks)												
UNIT 1	SECTION 2												
5	A manufacturer records the number of defective chips in a batch. Let X denote the number of defective chips. The probability distribution is: <table><tr><td>X</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>P(X)</td><td>k</td><td>2k</td><td>3k</td><td>4k</td><td>5k</td></tr></table> Find: a). k b). mean number of defectives c). variance defectives in the batch d). The cumulative distribution function F(x) e). $P(X \leq 2)$ Hint: $E(X) = \sum x P(X)$ $Var(X) = E(X^2) - [E(X)]^2$ (5 Marks)	X	0	1	2	3	4	P(X)	k	2k	3k	4k	5k
X	0	1	2	3	4								
P(X)	k	2k	3k	4k	5k								
6	The fraction of daily water consumption is represented by $f(x) = k x^2$, $0 < x < 1$. Calculate the following: 1. the value of k 2. average consumption of water on a day 3. variance in the consumption water on any two consecutive days Hint:												

$$\text{Mean} = E(x) = \int x f(x) dx$$

$$\text{Variance} = V(x) = E(x^2) - [E(x)]^2$$

(5 Marks)

7 The waiting time X (hours) of patients in a smart hospital has a probability density function:

$$f(x) = \begin{cases} ax, & 0 \leq x \leq 2 \\ a(4-x), & 2 \leq x \leq 4 \\ 0, & \text{otherwise} \end{cases}$$

Compute the following:

- 1). the parameter a
- 2). Cumulative density function of X

Hint:

$$\int_0^{\infty} f(x) dx = 1$$

and

$$\int_0^x f(x) dx = F(x)$$

(5 Marks)

8 (i) A teacher gives a quiz containing 5 multiple-choice questions. Let the random variable denote the number of questions answered correctly by a student. The probability distribution is given below.

x	0	1	2	3	4
P (x)	0.10	0.20	0.30	0.25	0.15

Find the following:

a). $P(X \geq 2)$

b). $P(1 < X \leq 4)$

(3 Marks)

(ii) A bookstore records the number of books purchased by customers during a promotional sale. A discrete random variable X has the following probability distribution:

X	0	1	2	3	4
P (x)	0.1	0.2	0.3	0.25	0.15

Find the mean of X.

Hint: $E(X) = \sum x P(X)$

(2 Marks)

UNIT 2

SECTION 1

9 A player successfully scores a free throw with probability 0.80. He attempts 10 free throws.

Find the probability of the following:

- a) Exactly 7 successful shots.

	b) All successful shots. c) At least 8 successful shots Hint: $P(X = x) = nC_x p^x q^{n-x}$ (5 Marks)
10	In a component manufacturing industry, there is a small probability of 1/500 for any component to be defective. The components are supplied in packets of 10. Use Poisson distribution to calculate the approximate number of packets containing a) No defective b) One defective component in a consignment of 10000 packets. Hint: $P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}$ (5 Marks)
11	There is a rainfall in a certain place is 10 days in every thirty days. Compute the probability that a) There is rainfall on at least 3 days of a given week. b) The first four days of a given week will be wet and the remaining days dry. Hint: $P(X = x) = nC_x p^x q^{n-x}$ (5 Marks)
12	(i) A football player, his success rate of goal hitting is 70%. What is the probability that player hits third goal on his fifth attempt? Hint: $P(X = x) = p(x) = (x - 1)C_{k-1} p^k q^{x-k}$ (3 Marks) (ii) Suppose a telemarketer has a 15% chance of making a sale on any given phone call. He is required to make 10 successful sales before leaving for the day. What is the probability that he needs to make more than 4 calls? Hint: $P(X = x) = p(x) = (x - 1)C_{k-1} p^k q^{x-k}$ (2 Marks)
UNIT 2	SECTION 2
13	Suppose that on an average 1 customer per minute arrives at a shop. Shopkeeper will wait until the customers arrive and the waiting time follows the gamma distribution. Calculate the probability that the shopkeeper will wait more than 5 minutes before the first customer arrive. Hint:

	$f(x) = \begin{cases} \frac{e^{-\lambda x} \lambda^\alpha x^{\alpha-1}}{\Gamma(\alpha)}, & \alpha, \lambda > 0, 0 < x < \infty \\ 0, & \text{elsewhere} \end{cases}$
	(5 Marks)
14	An average of one customer per minute arrive at a shop. Compute the probability that the shopkeeper will wait more than 5 minutes before both of the first two customers arrive. Hint: $f(x) = \begin{cases} \frac{e^{-\lambda x} \lambda^\alpha x^{\alpha-1}}{\Gamma(\alpha)}, & \alpha, \lambda > 0, 0 < x < \infty \\ 0, & \text{elsewhere} \end{cases}$
	(5 Marks)
15	A nationalized bank is facing a sex discrimination suit in which it is accused of paying men more than women. Based on this analysis the bankers collected the information from the large company about each employee. The annual salaries of employees in a large company are approximately normally distributed with a mean of \$50,000 and a standard deviation of \$20,000. a) Formulate the annual salaries of employees in the form of standard normal variate, if X be the normal variable denoting the annual salaries of employees. b) Compute the percent of people earning their annual salary less than \$40,000. Hint: $z = \frac{x - \mu}{\sigma} \text{ and } P(0 < Z < 0.5) = 0.1915$
	(5 Marks)
16	(i) An American household generates an average of 28 pounds of newspaper per month for garbage or recycling with the standard deviation of 2 pounds. If a household is selected at random, find the probability of generating the recycling less than 30 pounds per month. Hint: $z = \frac{x - \mu}{\sigma}, P(0 < Z < 1) = 0.3413$
	(3 Marks)
	(ii) A test score is normally distributed with a mean of 75 and a standard deviation of 5. Compute the Z - score for a test score of 85?
	(2 Marks)
UNIT 3	SECTION 1
17	An automobile manufacturer has developed a new engine design, Engine A, and claims it gives better fuel efficiency than the existing design, Engine B, used in its vehicles. Sample information: Engine A: $n_1 = 100$, $\bar{X}_1 = 18.5$ km/l, $\sigma_1 = 2.2$ km/l

	Engine B: $n_2 = 100$, $\bar{X}_2 = 17.6$ km/l, $\sigma_2 = 2.0$ km/l Test at the 5% significance level whether Engine A gives significantly higher average fuel efficiency than Engine B. Hint: $z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$, For 5% right-tailed test: Critical value = 1.645 (5 Marks)
18	A data centre operator runs two web servers, Server A and Server B, that host the same application, and wants to check whether Server A responds faster than Server B on average. Sample information: Server A: $n_1 = 90$, $\bar{X}_1 = 210$ ms, $\sigma_1 = 30$ ms Server B: $n_2 = 90$, $\bar{X}_2 = 240$ ms, $\sigma_2 = 35$ ms Test at the 5% significance level whether Server A has a significantly lower average response time than Server B. Hint: $z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$, For 5% left-tailed test: Critical value = -1.645 (5 Marks)
19	A structural engineer wants to compare the average tensile strength of welded steel joints produced by two welding techniques, Technique A and Technique B, before recommending one for a construction project. Technique A: $n_1 = 9$, $\bar{X}_1 = 410$ MPa, $S_1 = 15$ MPa Technique B: $n_2 = 11$, $\bar{X}_2 = 395$ MPa, $S_2 = 18$ MPa Test at the 5% level of significance whether the average tensile strengths of the two techniques differ significantly. Hint: $Sp^2 = \frac{[(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2]}{(n_1 + n_2 - 2)}$, $t = \frac{(\bar{x}_1 - \bar{x}_2)}{\left[Sp \times \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} \right]}$, df = $n_1 + n_2 - 2$, For 5% two-tailed test with v = 18: $t_{0.025, 18} = 2.101$ (5 Marks)
20	(i) A communication equipment manufacturer produces signal processing modules for mobile communication systems. To monitor product quality, engineers periodically collect samples from production and analyze the average output gain of the modules. The population mean output gain is known to be 60 dB and the population standard deviation is 12 dB. A random sample of 144 modules is selected for analysis. The sample of 144 modules gives an average output gain of 62 dB. Construct a 95% confidence

	interval for the population mean output gain. Hint: $CI = \bar{X} \pm 1.96 (\sigma / \sqrt{n})$, $SE(\bar{X}) = \sigma / \sqrt{n}$ (3 Marks)
	(ii) For the above sample, find the mean and standard error of the sampling distribution of the sample mean. (2 Marks)